

Aditya Pandey

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Software Engineer / Backend & Distributed Systems / AI-Powered Applications

Experience

Software Developer (Java, Spring Boot, Kafka, PostgreSQL)

National Australia Bank

Gurugram, India

Aug 2024 – Present

- Optimized database persistence by replacing JPA based save operations with JDBC native SQL queries, reducing bulk record insertion time from 8 minutes to 4 seconds by eliminating redundant duplicate-key checks during updates.
- Upgraded a legacy Java microservice from Java 11 to Java 17 and Spring Boot 2 to Spring Boot 3, remediating 25+ CVEs through Gradle dependency upgrades, and migrated AWS SDK v1 to v2 with asynchronous S3 transfers (TransferManager), improving system security and reducing I/O latency by 40%. (AWS S3, Gradle)
- Reduced test suite execution time by 50% by consolidating multiple WireMock instances into a singleton stub server for unified API mocking, resolving intermittent test failures. (JUnit, Mockito, WireMock)
- Engineered an event-driven Spring Boot microservice processing 10K+ daily Kafka events via real-time data transformations (Java Streams API) and idempotent writes to PostgreSQL, boosting data reliability and consistency across systems. (Kafka, PostgreSQL, Oracle, Java Streams)
- Solved high memory utilization during large dataset exports by introducing a Copy Manager that streams database results into 5MB chunks and uploads 10 chunks asynchronously to S3, limiting peak memory footprint to 50MB while reliably supporting large-scale exports.
- Implemented automated Kafka error handling by capturing and reprocessing failed messages through scheduled cron jobs, achieving 99.8% recovery success rate and ensuring message reliability.

Software Developer Intern

Buffercode (Internship Certificate)

Noida, India

June 2023 – Aug 2023

- Developed an interactive 3D web dashboard using React, Three.js, Framer Motion, and Context API — integrating Nivo charts with drag-and-drop functionality — enhancing user engagement by 20% through immersive animations and real-time visual insights.

Education

Delhi Technological University (Formerly DCE), Delhi

Bachelor of Technology in Computer Engineering

Dec 2020 – May 2024

GPA: 8.15

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering

Projects

Dialogus — AI-Powered Digital Media Platform

Next.js 15, TypeScript, Google

[GitHub](#) | [Live](#)

- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline that aggregates 50+ real-time news sources in parallel (Google News RSS, Serper API) with 5s abort timeouts, deduplicates using Jaccard similarity, and ranks by recency and keyword relevance to build grounded context for article generation.
- Engineered a multi-mode prompt system for Google Gemini with structured JSON output, supporting tiered grounding (strict with 3+ sources, lenient otherwise), zero-source fallback, and iterative regeneration with editor feedback — generating SEO-optimized, 500-word articles in under 15 seconds.
- Built a custom Markdown-to-Portable-Text parser and a multi-step publishing pipeline that uploads AI-generated cover images (Imagen 3.0) to Sanity CMS, resolves inline image placeholders, and publishes documents — secured end-to-end with HMAC-SHA256 session authentication and Next.js middleware.

DTU Estate

MongoDB, Express.js, React.js, Node.js, Tailwind CSS, Redux Toolkit, JWT, Firebase

[GitHub](#) | [Live](#)

- Engineered a MERN stack application with Redux Toolkit for state management and JWT authentication, serving 30+ daily active users and enabling secure user sessions via Google OAuth.
- Implemented CRUD operations for property listings with image uploads (Firebase Storage), advanced search filters, and a responsive UI — reducing housing search time by 40%.

Technical Skills

Languages: C++, Java, SQL, JavaScript/TypeScript

Backend & Data: Spring Boot, Kafka, Node.js/Express, PostgreSQL, MongoDB, JUnit, Mockito, WireMock

AI/LLM & Frontend: Google Gemini, RAG, Prompt Engineering, Imagen 3.0, Next.js, React, Tailwind CSS

DevOps & Tools: AWS S3, Docker, Harness CI/CD, Git, Sanity CMS